

St Catherine's School

Year: 12

Subject: Extension 1 Mathematics

Time allowed: 2 hours plus 5 minutes
reading time

Date: July 2006

Student number _____

Directions to candidates:

- All questions are to be attempted.(Q.1 to Q.7)
- Questions 1-3 are in booklet A.
- Questions 4-7 are in booklet B
- Each question is worth 12 marks
- Marks may be deducted for careless or badly arranged work

Marks:

Q 1	
Q 2	
Q 3	
Q 4	
Q.5	
Q.6	
Q.7	
Total	

Question 1

(12)

a) Prove the trigonometric identity $\operatorname{cosec} \theta - 2 \cot 2\theta \cos \theta = 2 \sin \theta$

(3)

b) Solve for x: $\frac{x^2 - 5}{x} > 4$

(3)

c) Find $\int \frac{1}{\sqrt{25 - 9x^2}} dx$

(3)

d) Find the general solution of the equation $2 \cos(4x + \frac{\pi}{3}) = \sqrt{2}$

(3)

Question 2 (start a new page)

(12)

- a) i) Write out the expansion of $(a+b)^n$ showing the first three terms, the general term, and the last term (1)

ii) Substituting appropriate values for a and b , show that $\sum_{k=0}^n (-1)^k {}^nC_k = 0$ (3)

b) Find the coefficient of x in the expansion of $(3x^2 - \frac{2}{x^3})^8$ (3)

c) i) Draw the graph of $y = \sin \frac{x}{2}$, $-2\pi \leq x \leq 2\pi$ (1)

ii) Use your graph to show that $\sin \frac{x}{2} + x + 1 = 0$ has only one solution. (1)

iii) Taking $x = -0.5$ as the first approximation to the solution, use one application of Newton's method to find a better approximation. (3)

Question 3 (start a new page)

(12)

- a) i) On the same set of axes, sketch the curve $f(x) = \log_e x$ and its inverse, $y = f^{-1}(x)$ (2)

ii) A (x,y) is a point on $y = \log_e x$
 B (y,x) is a point on $y = f^{-1}(x)$
 Plot A and B on your graph. (1)

iii) Show that the distance AB is $\sqrt{2} |(x - \log_e x)|$ (2)

iv) Find the minimum length of AB (3)

- b) Prove by Mathematical Induction that (4)

$$2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \dots + n \times 2^{n-1} = (n-1)2^n \quad \text{for integer } n \geq 2$$

Question 4 (start a new booklet)

(12)

a) Find $\int_0^{\frac{\pi}{4}} \cos^3 x \sin x dx$

(2)

b) Find $\int \frac{1}{(x^2 + 4)^2} dx$ using the substitution $x = 2 \tan \theta$

(3)

c) A polynomial $P(x)$ is given by

$$P(x) = ax^3 + bx^2 + 10x - 8$$

Find a and b if $(x + 2)$ is a factor of $P(x)$
and the remainder when $P(x)$ is divided by $(x - 1)$ is 12

(3)

d) α, β and γ are the roots of the equation $2x^3 - 4x - 7 = 0$

i) $\alpha^2 + \beta^2 + \gamma^2$

(2)

ii) $(\alpha + 1)(\beta + 1)(\gamma + 1)$

(2)

Question 5 (start a new page)

(12)

a) A rabbit population on a small island grows at a rate proportional to the difference between the population P and 100, i.e.

$$\frac{dP}{dt} = k(100 - P) \quad \text{where } t \text{ is measured in months}$$

i) Show that $P = 100 - Ae^{-kt}$ satisfies this condition.

(1)

ii) Initially the population is 6 rabbits, and after 2 months it has reached 20 rabbits.

Find values for A and k

(3)

iii) What is the expected number of rabbits on the island in the long term
(that is, as t becomes very large)?

(1)

b) i) Show that the curve $y = \sin x$ and the line $y = \frac{2x}{\pi}$ intersect

at the origin and at $(\frac{\pi}{2}, 1)$

(1)

ii) The region enclosed by the curve $y = \sin x$ and the line $y = \frac{2x}{\pi}$ is rotated about the x -axis to form a solid. Calculate the volume of the solid.

(3)

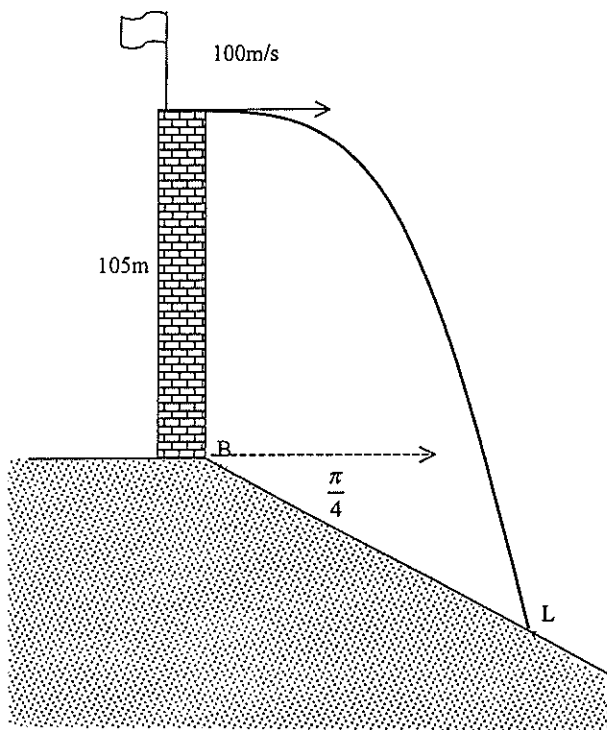
c) Graph the curve $y = 2 \sin^{-1} \frac{x}{3}$ showing clearly its domain and range

(3)

Question 6 (start a new page)

(12)

- a) A bullet is fired horizontally with a velocity of 100 m/s from the top of a tower 105 m high. The tower is at the top of a hill, which slopes downwards at an angle of depression of $\frac{\pi}{4}$. The bullet lands at L.



- i) Considering B, the base of the tower, as the origin, and using the acceleration due to gravity as -10 m/s^2 , show that the expressions for the x - and y - co-ordinates of the position of the bullet at time t sec are

$$x = 100t \quad \text{and} \quad y = 105 - 5t^2 \quad (2)$$

- ii) Show that the equation of the line BL is $y = -x$ (1)

- iii) Find the time taken for the bullet to hit the ground at L. (2)

- iv) Find the distance BL to the nearest metre. (1)

Question 6 (continued)

- b) i) Show that if $f(x)$ is an odd function defined for all x , then $f(0) = 0$ (1)

- ii) An odd polynomial $P(x)$ of degree 5 has a double zero at $x = 2$, and $P(1) = -12$ (2)

What is the leading term of $P(x)$?

- c) i) Find n if ${}^nC_{14} = {}^nC_{12}$ (1)

- ii) Simplify $\frac{{}^nC_r}{{}^nC_{r-1}}$ (2)

Question 7 (start a new page)

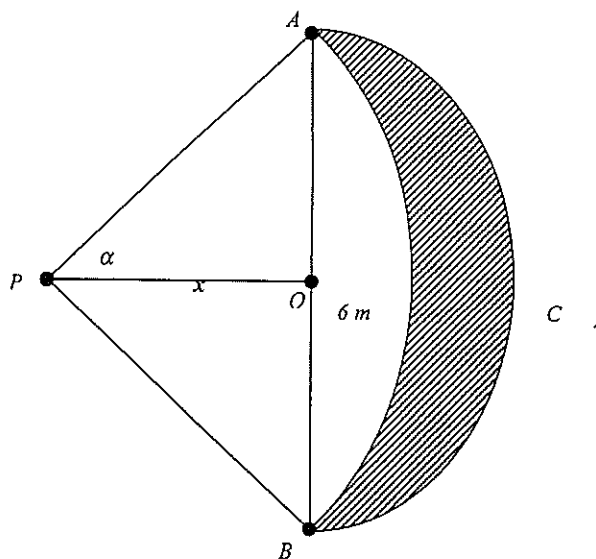
(12)

a) By considering the term in x^n on both sides of the identity $(1+x)^n(1+x)^n = (1+x)^{2n}$, show that

$$({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2 = {}^{2n}C_n$$

(3)

b)



A semicircle ACB has diameter AB 6 m long. O is the midpoint of AB. $OP \perp AB$. An arc of another circle, centre P, passes through A and B.

i) Show that if $OP = x$ m, then $\sin \alpha = \frac{3}{\sqrt{x^2 + 9}}$ (1)

ii) Show that the shaded portion S expressed as a function of x is (4)

$$S = \frac{9\pi}{2} + 3x - (x^2 + 9)\tan^{-1}\left(\frac{3}{x}\right)$$

iii) The point P moves to the left at 0.1 m/min. Find the rate of change of the area S when $x = 3$ m (3)

iv) Explain what happens to the shape of shaded area S as $x \rightarrow \infty$ (1)

SOLUTIONS

STUDENT NUMBER _____

COURSE
NAME

Sx Cath's
Maths Ext 1
Tmal 2006

SECTION

QUESTION

1-7

Q1?

a) $\cos 2\theta - 2 \cos \theta \sin \theta = 2 \sin \theta$

$$\begin{aligned} \text{LHS} &= \frac{1}{\sin \theta} - 2 \cos \theta \sin \theta \\ &= \frac{1}{\sin \theta} - \frac{2(\cos^2 \theta - \sin^2 \theta)}{2 \sin \theta \cos \theta} \end{aligned}$$

$$= \frac{1 - \cos^2 \theta + \sin^2 \theta}{\sin \theta}$$

$$= \frac{2 \sin^2 \theta}{\sin \theta}$$

$$= 2 \sin \theta$$

$$= \text{RHS as req.}$$

OTHER WAYS
POSSIBLE

(3)

b) $\frac{x^2 - 5}{x} > 4$

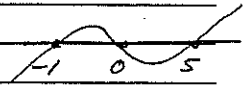
$$x(x^2 - 5) > 4x^2$$

$$x^3 - 4x^2 - 5x > 0$$

$$x(x^2 - 4x - 5) > 0$$

$$x(x - 5)(x + 1) > 0$$

$$-1 < x < 0, \quad x > 5$$



(3)

c) $\int \frac{1}{\sqrt{25 - 9x^2}} dx = \int \frac{1}{\sqrt{25 - 9x^2}} = \frac{1}{3} \int \frac{1}{\sqrt{\frac{25}{9} - x^2}}$

$$= \frac{1}{3} \sin^{-1} \left(\frac{3x}{5} \right) + C$$

d) $2 \cos(4x + \frac{\pi}{3}) = \sqrt{2}$

$$\cos(4x + \frac{\pi}{3}) = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$4x + \frac{\pi}{3} = 2\pi n \pm \frac{\pi}{4}$$

$$= \frac{2\pi n - \pi}{2} \quad \text{or} \quad \frac{\pi n - 7\pi}{4}$$

Q2

$$a) (a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} b + {}^nC_2 a^{n-2} b^2 + \dots + {}^nC_k a^{n-k} b^k + \dots + {}^nC_n b^n \quad (1)$$

b) Let $a=0, b=-1$

Then ${}^nC_0 + {}^nC_1(-1) + {}^nC_2(-1)^2 + \dots + {}^nC_k(-1)^k + \dots + {}^nC_n(-1)^n$
 $= (1-1)^n$

$$\therefore \sum_{k=0}^n {}^nC_k (-1)^k = 0 \quad \text{as } 1-1=0$$

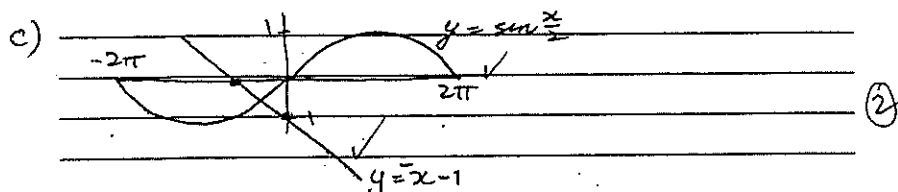
b) $(3x^2 + \frac{2}{x^3})^8 = (3x^2 + 2x^{-3})^8$

$$\Rightarrow {}^nC_k (3x^2)^{8-k} (-2x^{-3})^k \quad \text{Term } x^1$$

$$\therefore 2(8-k) - 3k = 1 \quad \checkmark$$

$$16 - 5k = 1 \quad \therefore k=3$$

$\therefore$ Coeff. is $-{}^8C_3 3^5 2^3 \quad \checkmark$

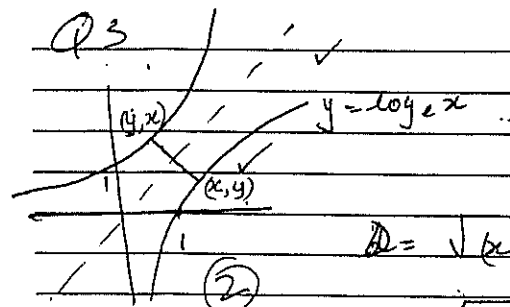


$$P(x) = \sin \frac{x}{2} + x + 1 \quad \checkmark \quad P(0.5) = 0.4794$$

$$P'(x) = \frac{1}{2} \cos \frac{x}{2} + 1 \quad \checkmark \quad P'(-0.5) = 1.48$$

$$x_2 = -0.5 - \frac{0.4794}{1.48} = -0.67$$

Q3



$$AB = \sqrt{(x-y)^2 + (y-x^2)^2} \quad \checkmark$$

$$= \sqrt{2(x-y)^2}$$

$$= \sqrt{2} |x-y|$$

$$= \sqrt{2} |x - \log_e x| \quad \checkmark$$

$$\frac{dD}{dx} = \sqrt{2} (1 - \frac{1}{x})$$

max/min $\frac{dD}{dx} = 0 \quad \therefore 1 - \frac{1}{x} = 0 \quad \therefore x=1$

$$\frac{d^2D}{dx^2} = \sqrt{2} (x^{-2}) > 0 \quad \text{for all } x \quad \therefore \text{min } D$$

$$AB = \sqrt{2} (1 - \log_e 1) = \sqrt{2} \quad \text{min length.}$$

b) Test for $n=2$.

$$\text{LHS} = 2 \times 2 = 4 \quad \text{RHS} = 1 \times 2^2 = 4 \quad \therefore \text{true for } n=2$$

Assume $\&$ for $n=k$

$$2 \times 2 + 3 \times 2^2 + \dots + k \times 2^{k-1} = (k-1) 2^k \quad \checkmark$$

Prove for $n=k+1$

$$2 \times 2 + 3 \times 2^2 + \dots + (k+1) 2^k = k \times 2^{k+1}$$

$$\text{LHS} = (k-1) 2^k + (k+1) 2^k$$

$$= 2k \times 2^k$$

$$= k 2^{k+1} = \text{RHS} \quad \checkmark$$

$\therefore$ true for $k+1$ if true for k .

$$\begin{aligned} \text{Q4} \quad \frac{\pi}{2} \int_0^{\frac{\pi}{4}} \cos^3 x \sin x \, dx &= \left[-\frac{1}{4} \cos^4 x \right]_0^{\frac{\pi}{4}} \\ &= -\frac{1}{4} \left(\frac{1}{\sqrt{2}} \right)^4 - \left(-\frac{1}{4} (1)^4 \right) \\ &= -\frac{1}{16} + \frac{1}{4} = \frac{3}{16} \end{aligned}$$

$$\text{b) } \int \frac{1}{(x^2+4)^{3/2}} dx \quad \begin{array}{l} x = 2 \tan \theta \\ \frac{dx}{d\theta} = 2 \sec^2 \theta \end{array}$$

$$\int \frac{1}{(4 \tan^2 \theta + 4)^{3/2}} \cdot 2 \sec^2 \theta \, d\theta$$

$$= \int \frac{1}{4^{3/2} (\sec^2 \theta)^{3/2}} \cdot 2 \sec^2 \theta \, d\theta$$

$$= \int \frac{1}{8 \sec^3 \theta} \cdot 2 \sec^2 \theta \, d\theta$$

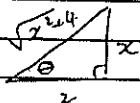
$$= \frac{1}{4} \int \frac{1}{\sec \theta} \, d\theta$$

$$= \frac{1}{4} \int \cos \theta \, d\theta$$

$$= \frac{1}{4} \sin \theta + C$$

$$= \frac{1}{4} \cdot \frac{x}{\sqrt{x^2+4}} + C$$

$$= \frac{x}{4\sqrt{x^2+4}} + C$$



4c)

$$P(x) = ax^3 + bx^2 + 10x - 8$$

$$P(-2) = 0 \quad \therefore -8a + 4b - 20 - 8 = 0$$

$$P(1) = 12 \quad a + b + 10 - 8 = 12$$

$$\therefore -8a + 4b = 28 \quad a + b = -14$$

$$-2a + b = 7$$

$$3a = -21$$

$$\therefore a = -7, b = -7$$

$$\text{d) } \alpha + \beta + \gamma = 0$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = -\frac{4}{2} = -2$$

$$\alpha\beta\gamma = \frac{7}{2}$$

$$\text{i) } \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= 0^2 - 2(-2) = 4$$

$$\text{ii) } (\alpha+1)(\beta+1)(\gamma+1) = \alpha\beta\gamma + (\alpha\beta + \beta\gamma + \gamma\alpha) + (\alpha + \beta + \gamma) + 1$$

$$= \frac{7}{2} + (-2) + 0 + 1$$

$$= \frac{5}{2}$$

Q5 $\frac{dP}{dt} = k(100 - P)$

(i) If $P = 100 - Ae^{-kt}$

LHS = $\frac{dP}{dt} = kAe^{-kt}$, RHS = $k(100 - P) = k(100 - 100 + Ae^{-kt}) = kAe^{-kt}$

$\therefore P = 100 - Ae^{-kt}$ satisfies $\frac{dP}{dt} = k(100 - P)$

(ii) at $t = 0$, $P = 6$

$6 = 100 - Ae^0 \therefore A = 94$

at $t = 2$, $P = 20$

$20 = 100 - 94e^{-2k}$

$\therefore e^{-2k} = \frac{-80}{-94}$

$k = -\frac{1}{2} \ln\left(\frac{80}{94}\right) \approx +0.0806$

(iii) as $t \rightarrow \infty$, $e^{-kt} \rightarrow 0 \therefore P \rightarrow 100$.
expected no is 100.

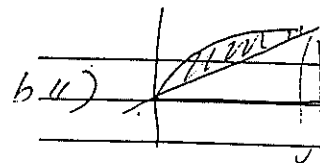
b) $y = \sin x$ $y = \frac{2x}{\pi}$

at $(0,0)$

$0 = \sin 0$ True $0 = \frac{2 \times 0}{\pi}$ True $\therefore (0,0)$ is intersect pt.

at $(\frac{\pi}{2}, 1)$

$1 = \sin \frac{\pi}{2}$ True $1 = \frac{2 \times \frac{\pi}{2}}{\pi}$ True $\therefore (\frac{\pi}{2}, 1)$ is intersect pt.



$\cos 2\theta = 1 - 2\sin^2 \theta$

$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$

$N = \pi \int_0^{\frac{\pi}{2}} y^2 dx$

OR $V = \frac{1}{2} \pi \int_0^{\frac{\pi}{2}} \left(\sin^2 x - \left(\frac{2x}{\pi} \right)^2 \right) dx$

$V = \pi \int_0^{\frac{\pi}{2}} \sin^2 x dx - \text{cone} = \frac{\pi^2}{4} - \frac{4}{\pi} \cdot \frac{\pi^3}{24}$
 $= \frac{\pi^2}{4} - \frac{\pi^2}{6}$
 $= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 2x) dx - \text{cone} = \frac{\pi^2}{12}$

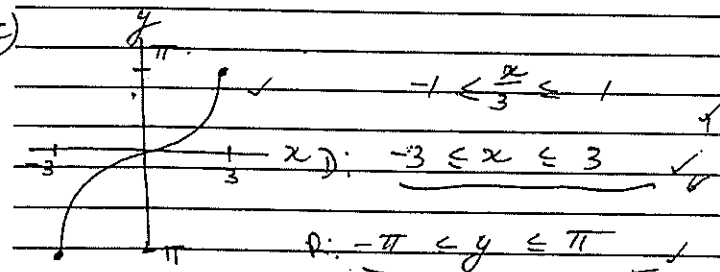
$= \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}} - \text{cone}$

$= \frac{\pi}{2} \left[\left(\frac{\pi}{2} - 0 \right) - (0 - 0) \right] - \text{cone} \quad V = \frac{1}{3} \pi r^2 h$

$= \frac{\pi^2}{4} - \frac{1}{3} \times \pi \times \frac{1}{2} \times \frac{\pi}{2}$

$= \frac{\pi^2}{4} - \frac{\pi^2}{6} = \frac{\pi^2}{12}$

c)



Question 6

$$a) i) \quad \ddot{x} = 0 \quad \ddot{y} = -10$$

$$\dot{x} = c_1 \quad \dot{y} = -10t + c_2$$

$$\text{at } t=0, \dot{x} = 100 \text{ and } \dot{y} = 0$$

$$\therefore c_1 = 100 \text{ and } c_2 = 0$$

$$\therefore \dot{x} = 100 \quad \dot{y} = -10t$$

$$x = 100t + c_3 \quad y = -5t^2 + c_4$$

$$\text{at } t=0, x=0 \text{ and } y=105$$

$$\therefore x = 100t \quad y = -5t^2 + 105$$

$$ii) \text{ Gradient of BL: } \tan\left(-\frac{\pi}{4}\right) = -1$$

BL passes through (0,0)

$$\therefore \text{eq is } y - 0 = -1(x - 0)$$

$$y = -x$$

$$iii) y = -x \quad x = 100t, \quad y = -5t^2 + 105$$

$$-5t^2 + 105 = -100t$$

$$5t^2 - 100t + 105 = 0$$

$$t^2 - 20t - 21 = 0$$

$$(t-21)(t+1) = 0$$

$$\therefore t = 21 \text{ or } t = -1 \text{ ignore -ve}$$

$\therefore$ takes 21 sec to hit ground

$$BL = \sqrt{(2100)^2 + (2100)^2}$$

$$= \sqrt{2} \times 2100$$

$$\approx 2970 \text{ m}$$

$$b) i) \quad f(x) \text{ is odd so } f(-x) = -f(x)$$

defined at 0 so $f(0) = -f(0)$

$$2f(0) = 0$$

$$\therefore f(0) = 0$$

$$ii) \quad P(x) = A(x-2)^2(x+2)^2 x$$

$$P(1) = -12 \text{ so } A(-1)^2(3)^2 \cdot 1 = -12$$

$$9A = -12$$

$$A = -4/3$$

$$\therefore \text{leading term is } -\frac{4}{3}x^5$$

$$c) i) \quad {}^nC_{14} = {}^nC_{12}$$

$$\text{so } r = 14$$

$$n - r = 12$$

$$\therefore n = 26$$

$$ii) \quad \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n!}{r!(n-r)!} \div \frac{n!}{(n-r+1)!(r-1)!}$$

$$= \frac{(n-r+1)(n-r)!}{r(n-r)!}$$

$$= \frac{n-r+1}{r}$$

Q7

$$a) \underline{LHS} = (1+x)^n (1+x)^n$$

$$= \left({}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_r x^r + \dots + {}^nC_n x^n \right) \left({}^nC_0 + {}^nC_1 x + \dots \right)$$

Term in x^n :

$${}^nC_0 \cdot {}^nC_n x^n + {}^nC_1 {}^nC_{n-1} x^n + {}^nC_2 {}^nC_{n-2} x^n + \dots$$

$$\dots + {}^nC_n {}^nC_0 x^n$$

But ${}^nC_r = {}^nC_{n-r}$ so ${}^nC_0 = {}^nC_n$ etc. ①

①

Coeff of x^n is

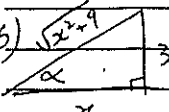
$$({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2$$

On RHS

Term in x^n is $2^n {}^nC_n x^n$ ①

equating co-eff on both sides

$$\therefore ({}^nC_0)^2 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2 = 2^n {}^nC_n$$

b)  by Pyth Rule, hypot is $\sqrt{x^2+9}$

$$\sin \alpha = \frac{3}{\sqrt{x^2+9}}$$

1) Shaded area =
Semicircle - (sector - triangle)

$$= \frac{\pi r^2}{2} - \frac{1}{2} r^2 \theta + \frac{1}{2} \times 6 \times x$$

$$= \frac{9\pi}{2} + 3x - \frac{1}{2} \times (\sqrt{x^2+9})^2 \cdot \theta$$

$$= \frac{9\pi}{2} + 3x - (x^2+9) \tan^{-1}\left(\frac{3}{x}\right)$$

$$\tan \alpha = \frac{3}{x} \quad \alpha = \tan^{-1} \frac{3}{x}$$

(i) Find $\frac{dS}{dx}$ by finding $\frac{dS}{dx} \times \frac{dx}{dx}$

$$\frac{dS}{dx} = 3 - \left[(x^2+9) \left(\frac{1}{1+\frac{9}{x^2}} \right) \cdot 3x^{-2} + \tan^{-1} \frac{3}{x} \cdot 2x \right]$$

$$= 3 - \left[(x^2+9) \frac{x^2}{x^2+9} \cdot \frac{-3}{x^2} + 2x \tan^{-1} \frac{3}{x} \right]$$

$$= 3 + 3 - 2x \tan^{-1} \frac{3}{x}$$

$$= 6 - 2x \tan^{-1} \frac{3}{x}$$

$$\frac{dS}{dx} = \frac{dS}{dx} \cdot \frac{dx}{dx}$$

$$= \left(6 - 2x \tan^{-1} \frac{3}{x} \right) (0.1)$$

$$\text{at } x=3 = \frac{6 - 2 \tan^{-1} 1}{10} = \frac{6 - \frac{3\pi}{4}}{10}$$

$$= \frac{12-3\pi}{20}$$

(iv) as $x \rightarrow \infty$, arc $\rightarrow AB$, $\therefore S$ app semicircle.